MATH 243 — Probability and Statistics

Measures of Dispersion: Range, Quartile and Mean Deviation, Variance, SD, CV

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Statistics portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — consistency / uniformity questions in almost every paper
Lesson 8 — Range, Quartile Deviation and Mean Deviation — absolute measures, empirical relations
Lesson 9 — Variance, Standard Deviation and Coefficient of Variation — grouped/ungrouped, CV comparisons, corrections
Quick Notes — Dispersion — revision bullets

NOTE Page numbers refer to 3. Measures of Dispersion.pdf (25 pp.), written as p. n. The slides use divisor N for the variance ($\sigma^2 = \Sigma(x - \bar{x})^2/N$); these notes follow them. "TF p." = merged question PDF page; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.3	Q5(c) Mean 60, SD 20 of 200 items; 3 and 67 taken instead of 13 and 17: corrected mean, SD, CV (12) [L9]
2022-23 (MATH 243) TF p. 49	Q5(c) Bulb lives, companies A and B (grouped): which company's bulbs to prefer (10) [L9]
2020-21 (MATH 245) TF p. 37	Q5(a) Daily wages of factories A and B (grouped): higher average wage; greater variability (10) [L9]
2019-20 (MATH 245) TF p. 34	Q5(a) Coefficient of variation, profits 20-30…90-100 (f 4, 8, 18, 30, 15, 10, 8, 7) (10) [L9]
2018-19 (MATH 245) TF p. 24	Q5(a) Shares A, B, C (average price, SD, current price): which is more stable; which to dispose of (15) [L9]
2015-16 (MATH 241) TF p. 15	Q6(b) Marks of students X and Y in 10 weeks: who is the better scorer, who is more consistent (17) [L9]
2013-14 (MATH 241) TF p. 7	Q6(a) Lives of refrigerator models A and B (grouped): which model has greater uniformity (15) [L9]
2011-12 (MATH 241) TF p. 2	Q6(b) Time study, samples A and B of 7 workers: which is more variable, which takes less time on average (15) [L9]

EXAM FOCUS 8 questions, 104 marks — a "which is more consistent / uniform / stable?" question in 8 of the 12 papers. The method never changes: mean and SD for each group, CV = $\sigma/\bar{x} \times 100$, the smaller CV is the more consistent. 2023-24 added the "wrongly copied items" correction. Master the grouped step-deviation table first — it is reused for skewness and moments.

Lesson 8 — Range, Quartile Deviation and Mean Deviation

pp. 1–8, 24–25

Dispersion measures how far individual values deviate from the central value. Batsman 1 (49, 50, 55, 54) and batsman 2 (10, 68, 90, 40) both average 52, but batsman 1 is far more consistent — the mean alone cannot show it (p. 1).

- **Absolute** measures are in the units of the data (range, QD, MD, variance, SD).
- **Relative** measures are ratios to an average, unit-free (coefficient of variation) — used to compare groups.

1. Range (p. 2)

R = L − S. Share prices 200–250: R = 250 − 160 = 90 Tk; grouped: upper limit of the last class − lower limit of the first (60 − 10 = 50). Limitation: depends only on the two extremes (three different series with R = 40).

2. Quartile Deviation (pp. 3–5)

$$\textbf{Q.D.} = (Q_3 - Q_1)/2$$

Profits of 100 companies: $Q_1 = 47.22$, $Q_3 = 70$ → middle 50 % lie in a range of 22.78 lakhs; Q.D. = 11.39 lakhs. Limitation: ignores the first and last 25 % of the data.

3. Mean Deviation (pp. 5–8)

$$\textbf{M.D.} = \Sigma|x - \bar{x}|/N \ \textbf{(ungrouped)}, \quad \Sigma f|x - \bar{x}|/N \ \textbf{(grouped)}$$

Batsman 1: $\Sigma|x - 52| = 3 + 2 + 3 + 2 = 10$ → M.D. = 2.5; batsman 2: 42 + 16 + 38 + 12 = 108 → 27. Sales (3, 6, 11, 3, 2): $\bar{x} = 33$, $\Sigma f|x - \bar{x}| = 204$ → M.D. = 8.16 lakhs. Marks (6, 5, 8, 15, 7, 6, 3): $\bar{x} = 33.4$, $\Sigma f|x - \bar{x}| = 659.2$ → 13.18. Practice: 6.3. Limitation: signs are ignored (otherwise the deviations sum to zero).

4. Empirical Relations (pp. 24–25)

$$\textbf{Q.D.} = (2/3)\sigma, \quad \textbf{M.D.} = (4/5)\sigma, \quad \textbf{Q.D.} = (5/6)\ \textbf{M.D.}$$

Profits (8, 12, 20, 6, 4): $\sigma = 11.14$ → M.D. ≈ 0.8 × 11.14 = 8.91. Marks example: M.D. 13.18 → Q.D. ≈ 10.98.

TRAP These relations hold only for (near) normal distributions; compute directly whenever the data are given.

Lesson 9 — Variance, Standard Deviation and Coefficient of Variation

pp. 9–23

$$\sigma^2 = \Sigma(x - \bar{x})^2/N = \Sigma x^2/N - \bar{x}^2$$
$$\text{Grouped: } \sigma^2 = [\Sigma fd^2/N - (\Sigma fd/N)^2] \times h^2, \quad \sigma = \sqrt{\sigma^2}$$
$$\text{C.V.} = (\sigma/\bar{x}) \times 100\ \%$$

Profit	Mid x	f	d = (x − A)/h	fd	fd^2
10–20	15	8	−2	−16	32
20–30	25	12	−1	−12	12
30–40	35	20	0	0	0
40–50	45	6	1	6	6
50–60	55	4	2	8	16
Total		N = 50		Σ = −14	Σ = 66

$\sigma^2 = [66/50 - (-14/50)^2] \times 10^2 = 1.2416 \times 100 = 124.16$; $\sigma = 11.143$; $\bar{x} = 35 + (-14/50)10 = 32.2$; C.V. = 34.6 % (pp. 10, 13, 20). Weekly wages of 10 workers: $\Sigma(x - 1323)^2 = 622 \rightarrow \sigma^2 = 62.2$, $\sigma = 7.89$ (pp. 9, 12). Rejects (boundaries 20.5–55.5, f 5, 15, 28, 42, 15, 12, 3): $\bar{x} = 36.96$, $\sigma = 6.74$, C.V. = 18.2 % (pp. 15, 22).

1. Coverage of $\bar{x} \pm k\sigma$ (pp. 16–17)

For a near-normal distribution $\bar{x} \pm \sigma$ covers 68.27 %, $\bar{x} \pm 2\sigma$ 95.45 %, $\bar{x} \pm 3\sigma$ 99.73 % of the observations. Breaking strength (f 3, 24, 27, 21, 5): $\bar{x} = 49.025$, $\sigma = 1.96 \rightarrow \bar{x} \pm 2\sigma = 45.1$ to 52.95 ≈ 45 to 53, containing about 76 of the 80 pieces (95 %).

2. Coefficient of Variation (pp. 17–23)

C.V. compares variability of data with different units or different means: height (40 in, sd 5) C.V. 12.5 % vs weight (10 kg, sd 2) 20 % → weight varies more; systolic 11.5 % vs diastolic 13.3 %. Share prices: Dhaka $\sigma = 8.33$, $\bar{x} = 115$, C.V. 7.24 %; Chittagong $\sigma = 10.09$, $\bar{x} = 119$, C.V. 8.48 % → Dhaka is safer.

MEMORIZE Smaller C.V. → more consistent / uniform / stable. Higher mean → better average (better scorer, longer life). If both differ, state each conclusion separately.

3. Correcting Wrong Items (textbook; asked 2023-24 Q5(c))

$$\text{Correct } \Sigma x = N\bar{x} - (\text{wrong values}) + (\text{right values})$$
$$\text{Correct } \Sigma x^2 = N(\sigma^2 + \bar{x}^2) - \Sigma(\text{wrong})^2 + \Sigma(\text{right})^2$$

Exam Approach

1. Grouped: mid-values, A (middle mid-value), d = (x − A)/h, columns f, fd, fd^2; totals.
2. $\bar{x} = A + (\Sigma fd/N)h$; $\sigma = h\sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)}$; C.V. = $\sigma/\bar{x} \times 100$.
3. Ungrouped: $\bar{x}$, then $\Sigma(x - \bar{x})^2$ in a table (or $\Sigma x^2 - N\bar{x}^2$); σ with divisor N.
4. Conclude in a sentence for each part of the question.

Practice — Lesson 9

(a) Actual — 2023-24 Q5(c) (22 batch p. 3). (b) Actual — 2022-23 Q5(c) (TF p. 49). (c) Actual — 2020-21 Q5(a) (TF p. 37). (d) Actual — 2019-20 Q5(a) (TF p. 34). (e) Actual — 2018-19 Q5(a) (TF p. 24). (f) Actual — 2015-16 Q6(b) (TF p. 15). (g) Actual — 2013-14 Q6(a) (TF p. 7). (h) Actual — 2011-12 Q6(b) (TF p. 2).

(a) The mean and SD of 200 items are 60 and 20. Two items were wrongly taken as 3 and 67 instead of 13 and 17. Find the correct mean, SD and coefficient of variation. (12)

(b) Bulb lives (hours) — Company A: 1700–1900: 10, 1900–2100: 16, 2100–2300: 20, 2300–2500: 8, 2500–2700: 6; Company B: 3, 40, 12, 3, 2. If prices are the same, which company's bulbs would you prefer? Justify. (10)

(c) Daily wages 20–30 … 80–90: Factory A 15, 30, 44, 60, 30, 14, 7; Factory B 25, 40, 60, 35, 20, 15, 5. Which factory pays higher average wages? Which has greater variability? (10)

(d) Coefficient of variation of profits 20–30 … 90–100 (f 4, 8, 18, 30, 15, 10, 8, 7). (10)

(e) Shares A, B, C: average price 18, 22.50, 24; SD 5.40, 4.50, 6.00; current price 36, 34.75, 39. (i) Which share is more stable? (ii) Holding all three, which would you dispose of now and why? (15)

(f) Marks in 10 weeks — X: 58, 59, 60, 54, 65, 66, 52, 75, 69, 52; Y: 87, 89, 78, 71, 73, 84, 65, 66, 56, 46. (i) Who is the better scorer? (ii) Who is more consistent? (17)

(g) Lives of refrigerators (years) 0–2 … 10–12 — Model A: 5, 16, 13, 7, 5, 4; Model B: 2, 7, 12, 19, 9, 1. Which model has greater uniformity? (15)

(h) Times (min) of 7 workers — A: 130, 124, 119, 134, 142, 130, 141; B: 132, 144, 136, 143, 132, 133, 140. Which sample is more variable and which takes less time on average? (15)

✓ Solution

(a)

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Wrong Σx = 200 × 60 = 12000;  correct Σx = 12000 − (3 + 67) + (13 + 17) = 11960  →  x̄ = 11960/200 = 59.8
Wrong Σx² = N(σ² + x̄²) = 200(400 + 3600) = 800000
Correct Σx² = 800000 − (3² + 67²) + (13² + 17²) = 800000 − 4498 + 458 = 795960
σ² = 795960/200 − 59.8² = 3979.8 − 3576.04 = 403.76  →  σ = 20.094
C.V. = 20.094/59.8 × 100 = 33.6 %
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(b)

Life (h)	Mid x	f	d = (x − A)/h	fd	fd^2
1700–1900	1800	10	−2	−20	40
1900–2100	2000	16	−1	−16	16
2100–2300	2200	20	0	0	0
2300–2500	2400	8	1	8	8
2500–2700	2600	6	2	12	24
Total		N = 60		Σ = −16	Σ = 88

A: $\bar{x} = 2200 + (-16/60) \times 200 = 2146.667$ h; $\sigma = 200 \times \sqrt{(88/60 - (-16/60)^2)} = 236.267$; C.V. = 11.01 %

Life (h)	Mid x	f	d = (x − A)/h	fd	fd^2
1700–1900	1800	3	−2	−6	12
1900–2100	2000	40	−1	−40	40
2100–2300	2200	12	0	0	0
2300–2500	2400	3	1	3	3
2500–2700	2600	2	2	4	8
Total		N = 60		Σ = −39	Σ = 63

B: $\bar{x} = 2200 + (-39/60) \times 200 = 2070$ h; $\sigma = 200 \times \sqrt{(63/60 - (-39/60)^2)} = 158.43$; C.V. = 7.65 %

A lasts longer on average (2146.7 h vs 2070 h) but B is more uniform (C.V. 7.65 % vs 11.01 %). At the same price prefer **A** for the longer average life; prefer B only if dependable, uniform life matters more than average life.

(c)

Wages	Mid x	f	d = (x − A)/h	fd	fd^2
20–30	25	15	−3	−45	135
30–40	35	30	−2	−60	120
40–50	45	44	−1	−44	44
50–60	55	60	0	0	0
60–70	65	30	1	30	30
70–80	75	14	2	28	56
80–90	85	7	3	21	63
Total		N = 200		Σ = −70	Σ = 448

A: $\bar{x} = 55 + (-70/200) \times 10 = 51.5$; $\sigma = 10 \times \sqrt{(448/200 - (-70/200)^2)} = 14.552$; C.V. = 28.26 %

Wages	Mid x	f	d = (x − A)/h	fd	fd^2
20–30	25	25	−3	−75	225
30–40	35	40	−2	−80	160
40–50	45	60	−1	−60	60
50–60	55	35	0	0	0
60–70	65	20	1	20	20
70–80	75	15	2	30	60
80–90	85	5	3	15	45
Total		N = 200		Σ = −150	Σ = 570

B: $\bar{x} = 55 + (-150/200) \times 10 = 47.5$; $\sigma = 10 \times \sqrt{(570/200 - (-150/200)^2)} = 15.124$; C.V. = 31.84 %

(i) Factory A pays the higher average wage (51.5 vs 47.5). (ii) Absolute variability: $\sigma_A = 14.55$, $\sigma_B = 15.12$; relative: C.V._A = 28.26 % < C.V._B = 31.84 % → Factory B has greater variability in wages.

(d)

Profits	Mid x	f	d = (x − A)/h	fd	fd^2
20–30	25	4	−4	−16	64
30–40	35	8	−3	−24	72
40–50	45	18	−2	−36	72
50–60	55	30	−1	−30	30
60–70	65	15	0	0	0
70–80	75	10	1	10	10
80–90	85	8	2	16	32
90–100	95	7	3	21	63
Total		N = 100		Σ = −59	Σ = 343

$\bar{x} = 65 + (-59/100)10 = 59.1$; $\sigma = 10\sqrt{(343/100 - 0.59^2)} = 17.555$; C.V. = 29.7 %.

(e) C.V. = SD/average × 100: A = 5.40/18 × 100 = 30 %; B = 4.50/22.50 × 100 = 20 %; C = 6/24 × 100 = 25 %.

(i) B has the smallest C.V., so its price is the most stable. (ii) Compare the current price with the average: A is 36/18 = 2.00 times its average, B 34.75/22.5 = 1.54, C 39/24 = 1.63. Share A is far above its average and also the least stable, so it is the one to sell now (its price is likely to fall back); keep B.

(f)

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X: Σx = 610, x̄ = 61.0, Σ(x − x̄)² = 526, σ = √(526/10) = 7.253, C.V. = 11.89 %
Y: Σx = 715, x̄ = 71.5, Σ(x − x̄)² = 1710.5, σ = √(1710.5/10) = 13.079, C.V. = 18.29 %
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(i) Y is the better scorer (higher mean 71.5). (ii) X is more consistent (smaller C.V. 11.89 % < 18.29 %).

(g)

Life (yr)	Mid x	f	d = (x − A)/h	fd	fd^2
0–2	1	5	−3	−15	45
2–4	3	16	−2	−32	64
4–6	5	13	−1	−13	13
6–8	7	7	0	0	0
8–10	9	5	1	5	5
10–12	11	4	2	8	16
Total		N = 50		Σ = −47	Σ = 143

A: $\bar{x} = 7 + (-47/50) \times 2 = 5.12$ yr; $\sigma = 2 \times \sqrt{(143/50 - (-47/50)^2)} = 2.812$; C.V. = 54.92 %

Life (yr)	Mid x	f	d = (x − A)/h	fd	fd^2
0–2	1	2	−3	−6	18
2–4	3	7	−2	−14	28
4–6	5	12	−1	−12	12
6–8	7	19	0	0	0
8–10	9	9	1	9	9
10–12	11	1	2	2	4
Total		N = 50		Σ = −21	Σ = 71

B: $\bar{x} = 7 + (-21/50) \times 2 = 6.16$ yr; $\sigma = 2 \times \sqrt{(71/50 - (-21/50)^2)} = 2.23$; C.V. = 36.21 %

C.V._B = 36.21 % < C.V._A = 54.92 % → Model B has greater uniformity (and also a longer mean life).

(h)

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A: x̄ = 920/7 = 131.429, Σ(x − x̄)² = 423.714, σ = 7.78, C.V. = 5.92 %
B: x̄ = 960/7 = 137.143, Σ(x − x̄)² = 160.857, σ = 4.794, C.V. = 3.5 %
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Sample A is more variable (C.V. 5.92 % > 3.50 %) and takes less time on average (131.4 < 137.1 minutes).

Quick Notes — Dispersion

Revision points for Lessons 8–9.

L8 · Range, QD, MD

- R = L − S; Q.D. = (Q3 − Q1)/2 (profits 11.39); M.D. = $\Sigma f|x - \bar{x}|/N$.
- Empirical: Q.D. = 2σ/3, M.D. = 4σ/5, Q.D. = 5 M.D./6.

L9 · SD and CV

- $\sigma = h\sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)}$ (divisor N); C.V. = $\sigma/\bar{x} \times 100$; smaller C.V. = more consistent.
- Corrections: fix Σx and Σx^2 then recompute (59.8, 20.09, 33.6 %).
- Bulbs: A mean 2146.7 C.V. 11.0 %, B 2070 C.V. 7.65 %. Wages: A higher mean, B more variable.
- Shares: CV 30/20/25 % → B stable, sell A. Fridges: B more uniform. Marks: Y better, X consistent.